IEOR 221

Homework 7

Module 7 & 8 Options – BSM Eqn, Greeks & Volatility (October 22, 2025)

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1) Assume that a non-dividend-paying stock has an expected return of μ and a volatility of σ. An innovative financial institution has just announced that it will trade a security that pays off a dollar amount equal to $\ln S_T$ at time T, where S_T denotes the value of the stock price at time T.
(a) Use risk-neutral valuation to calculate the price of the security at time t in terms of the stock price, S, at time t. The risk-free rate is r.
(b) Confirm that your price satisfies the BSM differential equation.
(Hull 15.19)

2) If the volatility of a stock is 18% per annum, estimate the standard deviation of the percentage price change in (a) 1 day, (b) 1 week, and (c) 1 month. (Hull 15.34)
(assume there are 52 weeks and 250 business days in a year)

3) The spot price of an index is 1,000 and the risk-free rate is 4%. The prices of 3-month European call and put options when the strike price is 950 are 78 and 26. Estimate
(a) the dividend yield and (b) the implied volatility. (Hull 17.30)

4) A European call option on a certain stock has a strike price of $30, a time to maturity of 1 year, and an implied volatility of 30%. A European put option on the same stock has a strike price of $30, a time to maturity of 1 year, and an implied volatility of 33%. What is the arbitrage opportunity open to a trader? Does the arbitrage work only when the lognormal assumption underlying Black–Scholes–Merton holds? Explain carefully the reasons for your answer. (Hull 20.17)